

TED (15) 3013

THIRD SEMESTER DIPLOMA EXAMINATION IN CIVIL
ENGINEERING/ TECHNOLOGY

Reg. No.....

(MODEL QUESTION PAPER)

Signature.....

THEORY OF STRUCTURES – I

(Common for CE, AR, QS, EV and WR)

[Time : 3 hours

(Maximum marks :100)

PART—A

(Maximum marks : 10)

Marks

I Answer the following questions in one or two sentences. Each questions carries 2 marks.

1. List any two elements of force
2. Define the term moment of inertia.
3. State Poisson's ratio.
4. Define the term torque.
5. Write down the bending equation
6. (5 x 2 =10)

PART B

II. Answer any five question from the following , Each question carries 6 marks

1. Calculate the support reactions of a simply supported beam AB of 5m span with a point Load of 20 kN at 2m from left support A and a u.d.l of 3kN/m for a length of 3m from right support B .
2. Find the centre of gravity of a 'T' section with flange 150mm x 10mm and web also 150mm x 10mm.
3. A steel rod of 40mm diameter and 4m length is subjected to an axial pull of 50KN. Find the stress, strain and elongation of the rod. Take $E = 2 \times 10^5 \text{ N/mm}^2$
4. Draw the SFD and BMD of a cantilever beam AB of span 2.5m with a u.d.l of 1kN/m along its whole span.
5. A circular shaft 30mm diameter is subjected a torque of 20 KNm. Calculate the maximum shear stress developed in the shaft
6. Two parallel walls 5m apart are stayed together by a steel rod 15mm diameter passing through metal plates and nuts at each end . The nuts are tightend when the rod is at a

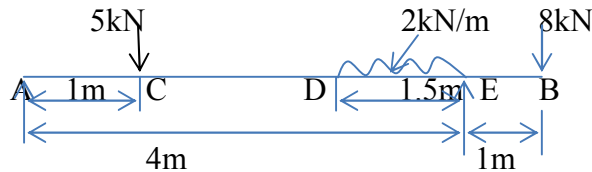
temperature of 80° . Determine the stress in the rod when the temperature falls down to 60° . Take $E=200\text{GPa}$ and $\alpha=12\times10^{-6}/^\circ\text{C}$.

1. A rectangular beam 60mm wide and 120mm deep is subjected to a shearing force of 50kN. Calculate the maximum shear stress and draw the distribution diagram for the shear stress. (5 x 6 = 30)

PART C MODULE – I

Answer one full question from each module. Each full question carries 15 marks

III(a) Calculate the support reactions of the overhanging beam as shown in fig.



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- (b) Locate the centroid of a circle and Triangle

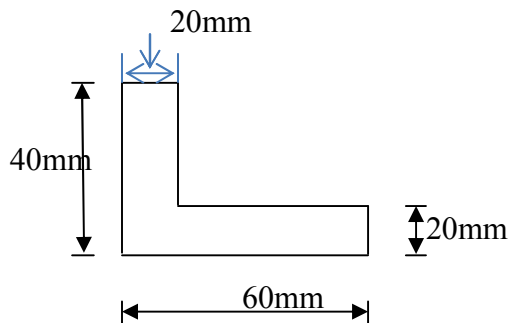
7

OR

- (a) A simply supported beam AB of 6m span is subjected to a u.d.l of 1kN/m for a distance 2m from left support A and a point load of 6kN at a point 2m from right support B. calculate the moment of forces at a section 'C' at a distance 3m from A.

8

- (b) Calculate the moment of inertia of the 'L' section shown in fig about a vertical axis passing through its C.G.



7

MODULE II

- (a) A prismatic metallic bar of rectangular section 400mm x 250mm and 2m long is subjected to a load of 100kN applied gradually on it. If the stress at elastic limit of bar material is 200 N/mm^2 . Determine (a) strain energy at the given load (b) Proof resilience (c) Modulus of resilience, Take $E = 200\text{ KN/mm}^2$.

8

(b) A bar of 30mm diameter is subjected to a pull of 70kN. The measured extension on a gauge length of 200mm is 0.12mm and the change in diameter is 0.004mm. Calculate Poisson's ratio and value of three moduli. 7

OR

VI (a) A concrete column of 340mm diameter is reinforced with 4 bars of 20mm diameter. Find the stress in steel when the concrete is subjected to a stress of 4.5MPa. Also find the safe load the column can carry. Take $E_s/E_c = 18$ 8

(b) Compute the deformation of a body due to its own weight. 7

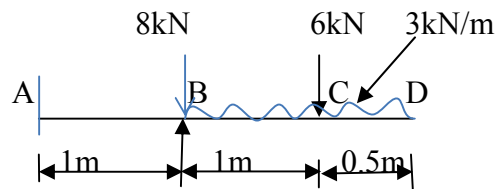
MODULE – III

VII (a) A simply supported beam having length 6m, it carrying a uniformly distributed load of 3kN/m over a length of 1.5m from the left end and a udl of 2kN/m over a length of 3m from the right end and also a point load of 4kN at a distance of 1.5m from the right end. Draw the SFD and BMD for the beam and find the position and value of maximum bending moment. 8

(b) The maximum Torque transmitted by a shaft is 3000Nm. If the allowable shear stress in the shaft is 40N/mm², determine the diameter of the shaft. 7

OR

VIII (a) Sketch SFD and BMD of a cantilever beam shown in fig.



8

(b) A steam boiler 3m diameter is required to carry a net working pressure of 1 N/mm² stress in the metal should not exceed 80N/mm². Find the thickness of the metal if efficiency of the joint is 70% 7

MODULE IV

IX (a) A rectangular beam 300mm deep and 400mm wide is simply supported over a span of 8m, find the udl per meter run the beam can carry, if bending stress is not to exceed 120 N/mm². 8

(b) Sketch the shear stress distribution diagram for a symmetrical I- section of flange width 'B', web width 'b', over all depth 'D', thickness of flange 't' and depth of web 'd'. Mark the shear stress at different levels of section. 7

OR

X (a) Derive the equation in the theory of simple bending 8

(b) The average shear stress at the section of a simply supported rectangular beam of size 120mmx250mm is 0.6 N/mm^2 . Determine the shear force at the section and the maximum shear stress at the section. 7